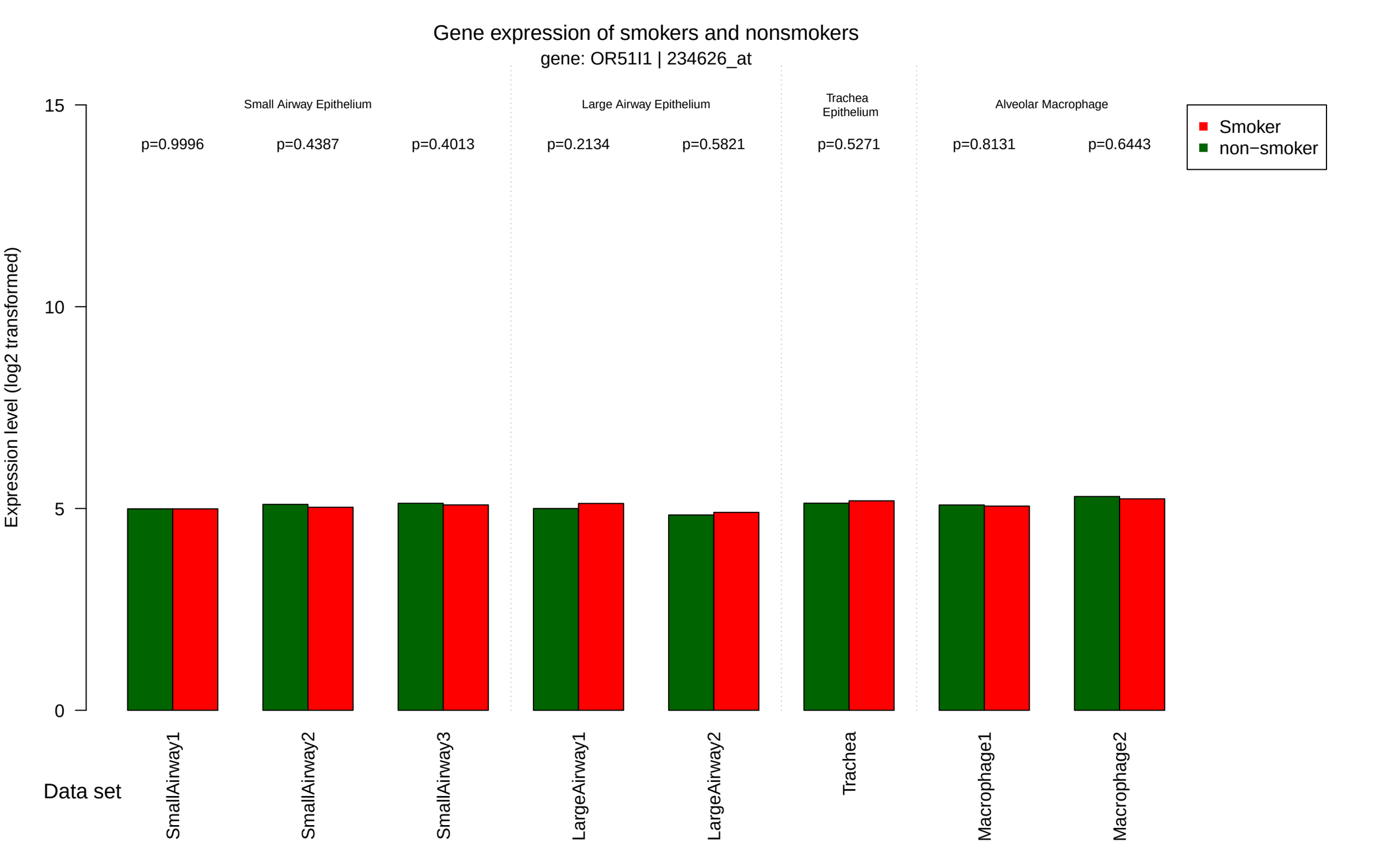
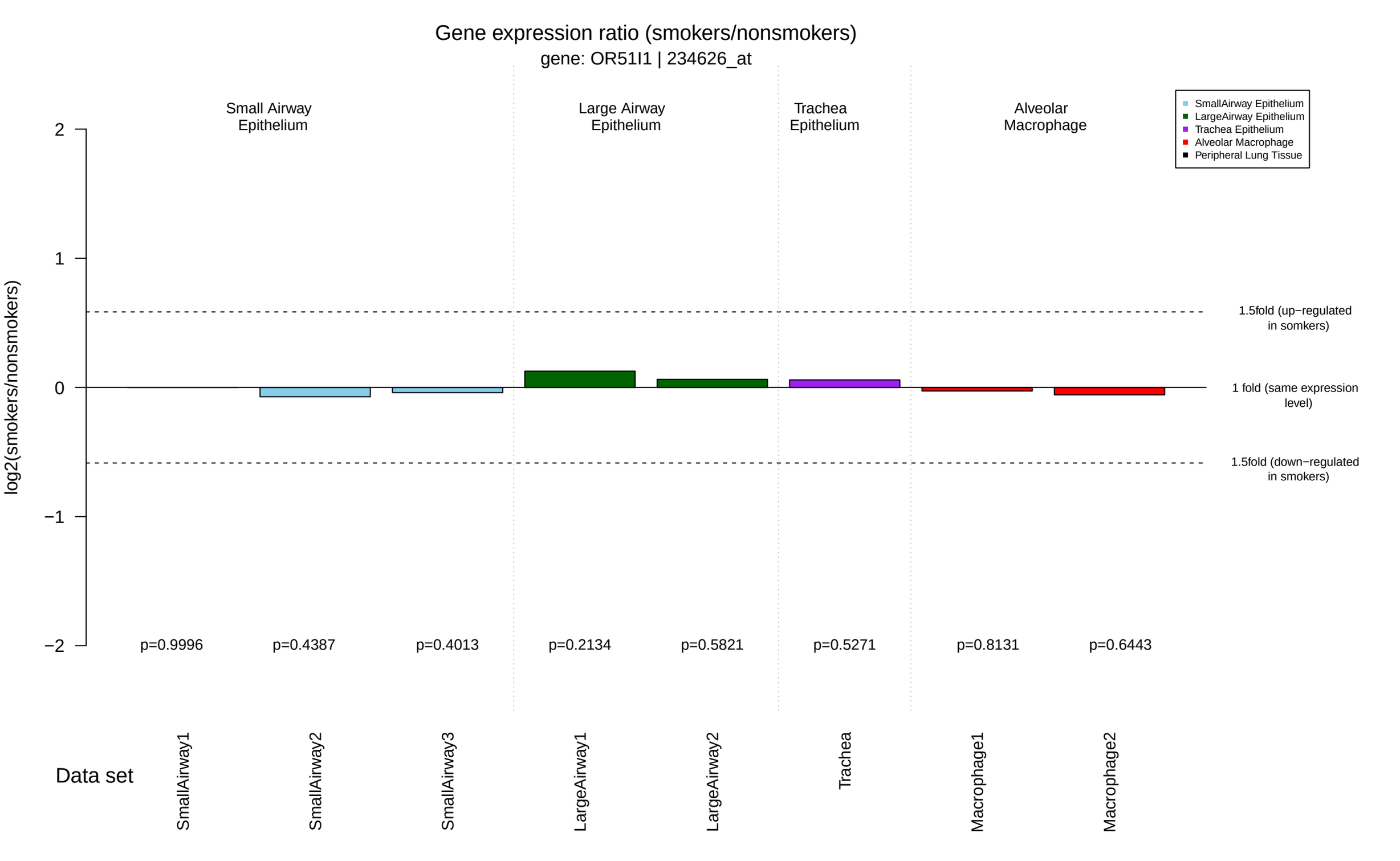
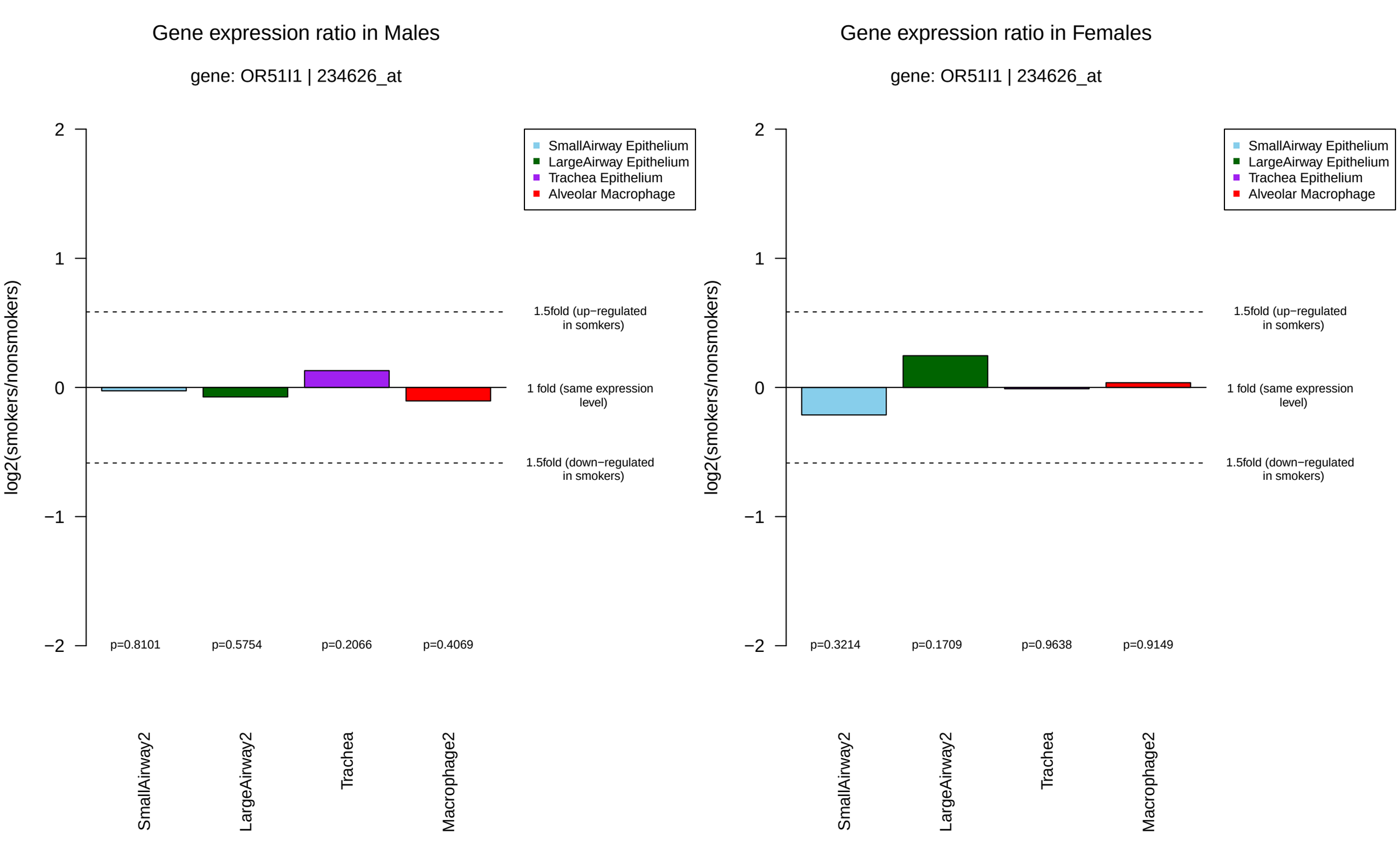
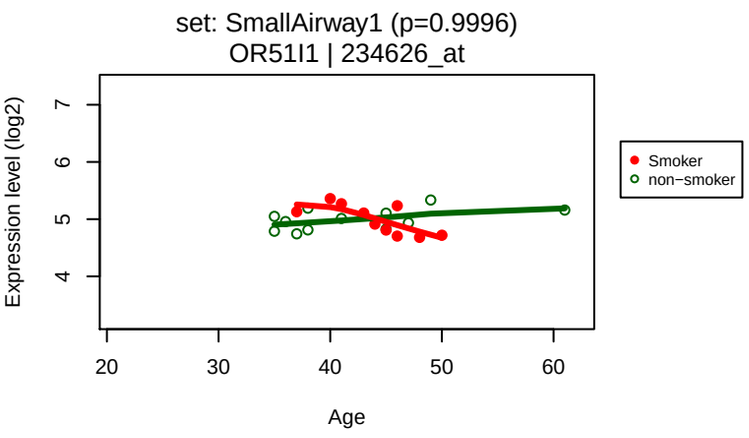
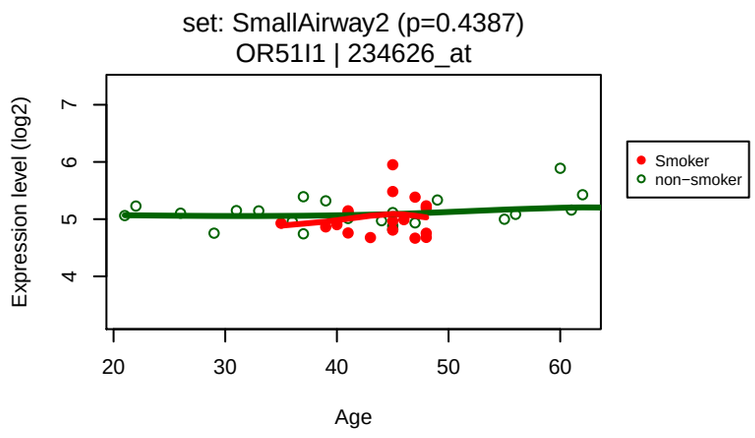
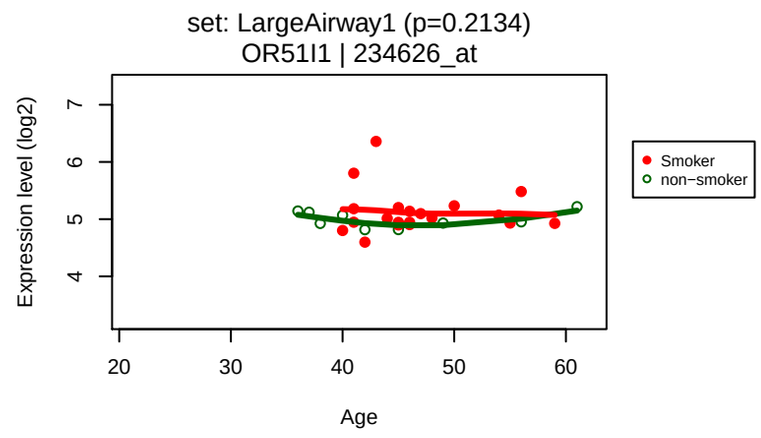

Gene expression of smokers and nonsmokers
gene: OR51I1 | 234626_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9996
p=0.4387
p=0.4013
p=0.2134
p=0.5821
p=0.5271
p=0.8131
p=0.6443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

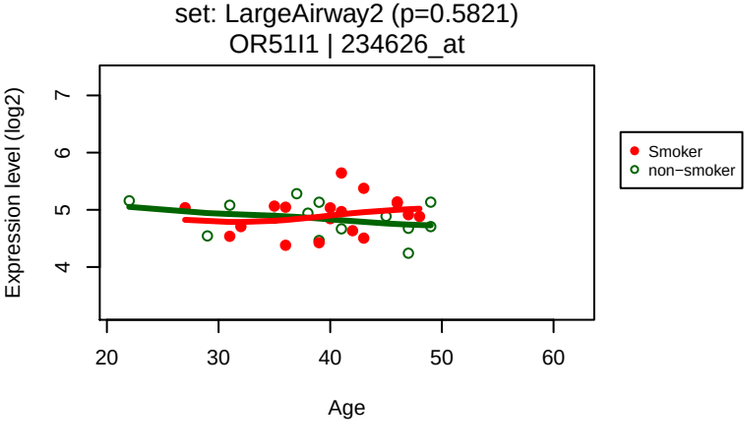

Gene expression ratio (smokers/nonsmokers)
gene: OR51I1 | 234626_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9996
p=0.4387
p=0.4013
p=0.2134
p=0.5821
p=0.5271
p=0.8131
p=0.6443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

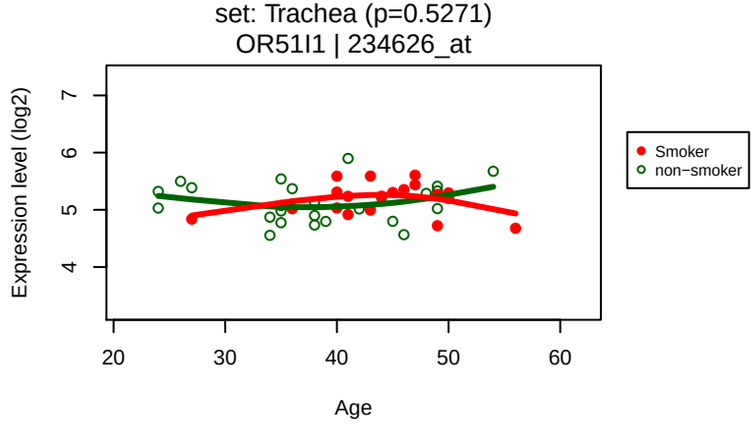

Gene expression ratio in Males
gene: OR51I1 | 234626_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8101
p=0.5754
p=0.2066
p=0.4069
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51I1 | 234626_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3214
p=0.1709
p=0.9638
p=0.9149
SmallAirway2
LargeAirway2
Trachea
Macrophage2

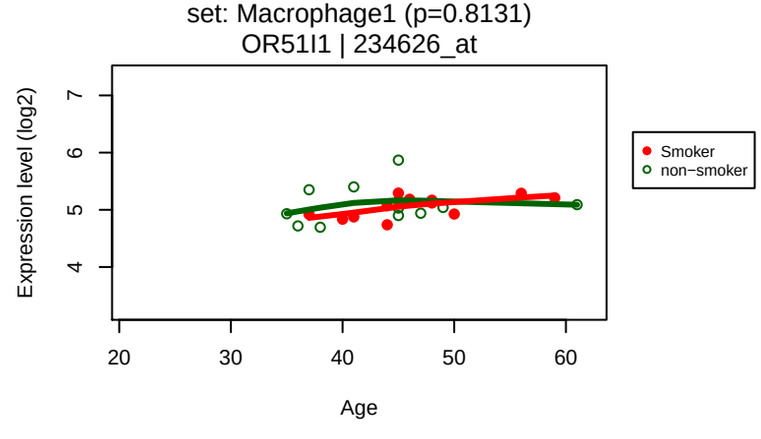

set: SmallAirway1 (p=0.9996)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker

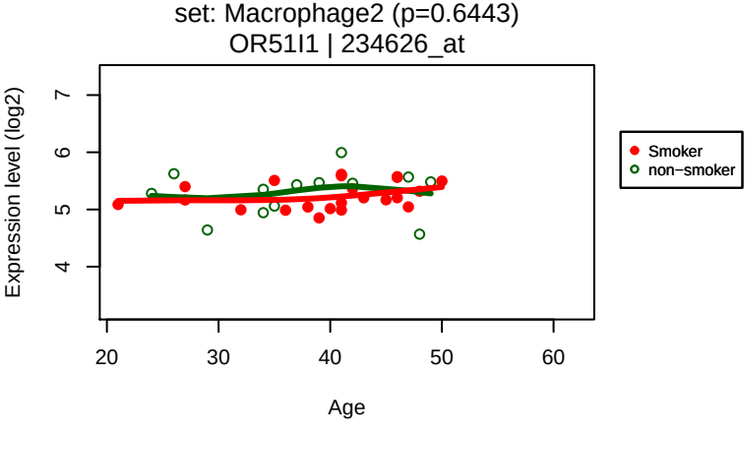

set: SmallAirway2 (p=0.4387)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker

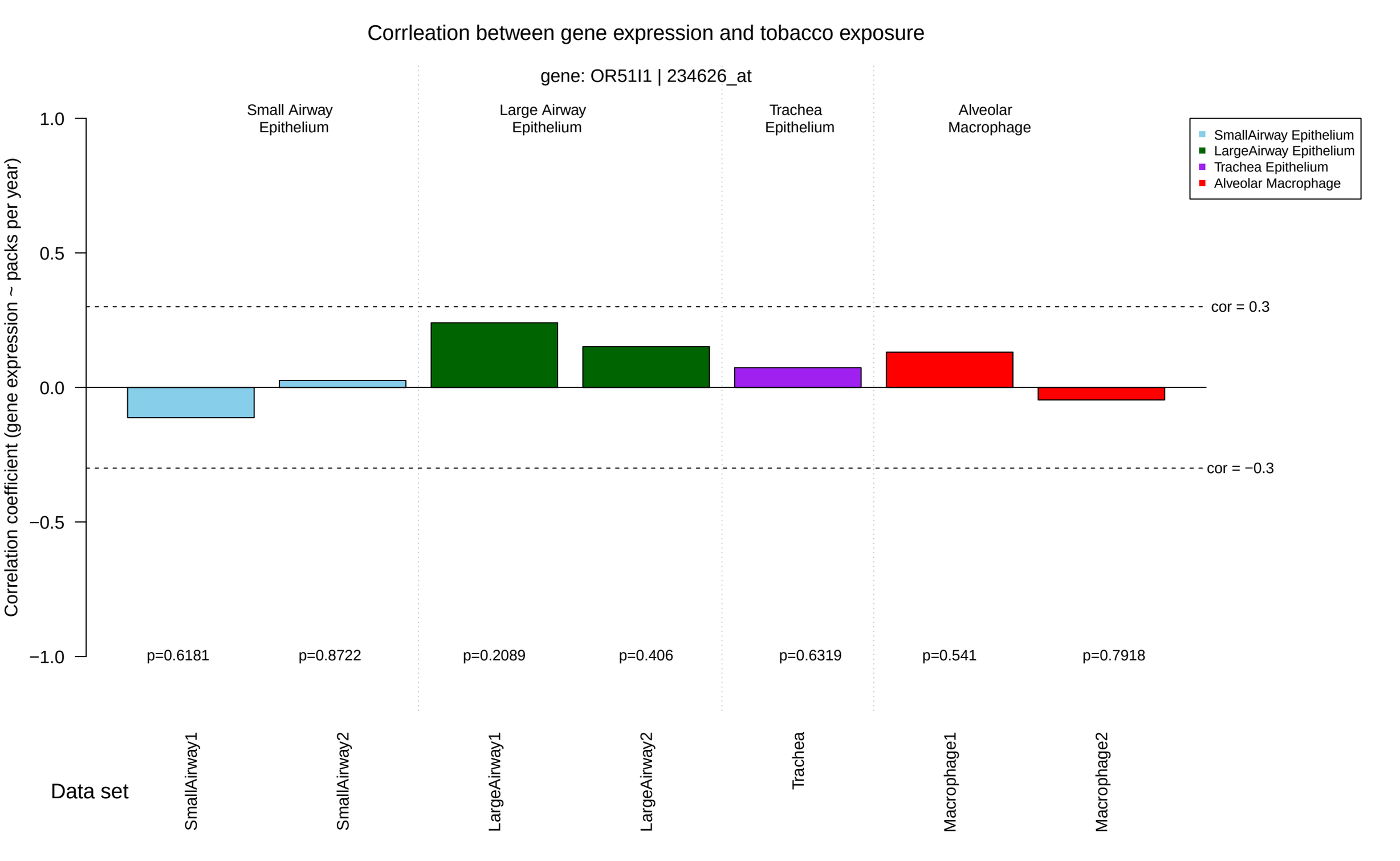

set: LargeAirway1 (p=0.2134)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5821)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5271)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8131)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6443)
OR51I1 | 234626_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51I1 | 234626_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6181
p=0.8722
p=0.2089
p=0.406
p=0.6319
p=0.541
p=0.7918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2